arXiv:2412.10427v2 [cs.CL] 10 Jan 2025

Identifying and Manipulating Personality Traits in LLMs Through Activation Engineering

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Abstract

The field of large language models (LLMs) has grown rapidly in recent years, driven by the desire for better efficiency, interpretability, and safe use. Building on the novel approach of "activation engineering," this study explores personality modification in LLMs, drawing inspiration from research like "Refusal in LLMs is Mediated by a Single Direction" [1] and "Steering Llama 2 via Contrastive Activation Addition." [8] We leverage activation engineering to develop a method for identifying and adjusting activation directions related to personality traits, which may allow for dynamic LLM personality fine-tuning. This work aims to further our understanding of LLM interpretability while examining the ethical implications of such developments.

1 Introduction

Large language models have been developed with ongoing efforts to improve their functionality, comprehend their internal workings, and guarantee their ethical and safe application. New developments in the field have led to the concept of *'activation engineering'*[13], which posits that activation vectors can mediate particular behaviors within LLMs. This development has made it possible to adjust and regulate the output of these models in new ways.

This paper is motivated by the potential to extend this line of inquiry into the domain of personality traits in LLMs. The ability to dynamically adjust the personality of a language model without extensive retraining could mark a significant advancement in the field, offering improved flexibility in AI applications. This approach could potentially revolutionize how we interact with and deploy AI systems, allowing for more personalized and context-appropriate responses.

However, the pursuit of such capabilities is not without challenges. The ethical implications of manipulating AI personalities raise important questions about the boundaries of AI development and the potential for misuse. By focusing on the intersection of activation engineering and personality manipulation, our research aims to contribute to the broader discourse on AI ethics.

2 Methods

2.1 Feature Induction via Weight Orthogonalization

TL;DR Methodology We propose an approach for 'fine-tuning' certain personality features in large language models (LLMs) that we name "feature induction." By comparing the activation vectors generated by prompts expressing a desirable personality trait to those generated by neutral prompts, we can find a direction in the activation space that correlates to that feature. This approach allows for precise personality changes without affecting the model's overall understanding or necessitating extensive retraining.[7]

2.2 Methodology

Our method is an adaptation of *feature ablation via weight orthogonalization*[1]; however, our goal is to induce and increase desired personality traits in the responses of the model, rather than suppressing undesirable features. We use a *uncensored language model* to examine the widest variety of personalities, including those that are frequently considered sensitive or improper. With this decision, we can investigate features that could be intentionally repressed in safety-mechanism-equipped, commercial versions.

2.2.1 Original Concept: Feature Ablation

The basic approach to feature ablation through weight orthogonalization is to intervene in the model's inference phase to stop it from representing a particular direction in its activation space, which is represented by the notation $\hat{r}$. This undesirable direction frequently reflects a behavior or characteristic that the model developers are attempting to suppress. For every contribution $\mathbf{c}_{\text{out}} \in \mathbb{R}^{d_{\text{model}}}$ to the residual stream, the component along the $\hat{r}$ direction is effectively "zeroed out" using the following operation:

$$\mathbf{c}'_{\text{out}} \leftarrow \mathbf{c}_{\text{out}} - (\hat{r} \cdot \mathbf{c}_{\text{out}})\hat{r} \tag{1}$$

Direct modifications to the weight matrices of the model can also accomplish this $\hat{r}$ component elimination. To make sure that information is never written along a particular direction, $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{input}}}$ can be orthogonalized with regard to each matrix that contributes to the residual stream:

$$\mathbf{W}'_{\text{out}} \leftarrow \mathbf{W}_{\text{out}} - (\hat{r} \cdot \mathbf{W}_{\text{out}})\hat{r} \tag{2}$$

The attention output matrices, MLP output matrices, positional embedding matrix, and embedding matrix are among the matrices in a transformer architecture that write to the residual stream. We essentially stop the model from expressing or activating that direction by orthogonalizing all of these matrices related to that direction $\hat{r}$, which suppresses the related behavior or feature.

2.2.2 Our Implementation: Feature Induction

Instead of suppressing features, we leverage a similar principle to enhance desired personality traits.

1. Calculate the Difference in Means for Activations The mean activation vectors for two sets of prompts—those that match the desired personality trait and those that don't—are first calculated. The direction connected to the personality trait is indicated by the difference between these mean vectors.

We extract the activation vectors for each prompt from a particular layer in the architecture of the model. We discovered through empirical observation that Layer 18 had the biggest impact on how personality traits were expressed in the model's output.[2]

For both the *neutral* and *trait* prompts, we compute the mean activation vector. We obtain the *personality direction vector* $\mathbf{r}$ from the difference between these two mean vectors. In essence, this vector indicates the area of the activation space where the target personality trait is most strongly expressed:

$$\mathbf{r} = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{a}_i^{\text{trait}} - \frac{1}{n_n} \sum_{i=1}^{n_n} \mathbf{a}_i^{\text{neutral}} \tag{3}$$

where:

- $\mathbf{r}$ is the personality direction vector.
- n_t is the number of samples with the desired personality trait.
- n_n is the number of neutral samples (without the trait).
- $\mathbf{a}_i^{\text{trait}}$ is the activation vector for the i-th sample with the desired personality trait.
- $\mathbf{a}_i^{\text{neutral}}$ is the activation vector for the i-th neutral sample.

An alternative approach involves computing the difference between each pair of corresponding *trait* and *neutral* activation vectors and then calculating the mean of these differences. This approach can potentially capture more nuanced variations in the data: